

OpenStax Calculus Volume 2

Section 5.1 Exercises: Sequences

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Find the first six terms of each of the following sequences, starting with $n = 1$.

1. $a_n = 1 + (-1)^n$ for $n \geq 1$
2. $a_n = n^2 - 1$ for $n \geq 1$
3. $a_1 = 1$ and $a_n = a_{n-1} + n$ for $n \geq 2$
4. $a_1 = 1$, $a_2 = 1$ and $a_{n+2} = a_n + a_{n+1}$ for $n \geq 1$
5. Find an explicit formula for a_n where $a_1 = 1$ and $a_n = a_{n-1} + n$ for $n \geq 2$.
6. Find a formula a_n for the n th term of the arithmetic sequence whose first term is $a_1 = 1$ such that $a_{n+1} - a_n = 17$ for $n \geq 1$.
7. Find a formula a_n for the n th term of the arithmetic sequence whose first term is $a_1 = -3$ such that $a_{n+1} - a_n = 4$ for $n \geq 1$.
8. Find a formula a_n for the n th term of the geometric sequence whose first term is $a_1 = 1$ such that $\frac{a_{n+1}}{a_n} = 10$ for $n \geq 1$.
9. Find a formula a_n for the n th term of the geometric sequence whose first term is $a_1 = 3$ such that $\frac{a_{n+1}}{a_n} = 1/10$ for $n \geq 1$.
10. Find an explicit formula for the n th term of the sequence whose first several terms are $\{0, 3, 8, 15, 24, 35, 48, 63, 80, 99, \dots\}$. (Hint: First add one to each term.)
11. Find an explicit formula for the n th term of the sequence satisfying $a_1 = 0$ and $a_n = 2a_{n-1} + 1$ for $n \geq 2$.

Find a formula for the general term a_n of each of the following sequences.

12. $\{1, 0, -1, 0, 1, 0, -1, 0, \dots\}$ (Hint: Find where $\sin x$ takes these values)
13. $\{1, -1/3, 1/5, -1/7, \dots\}$

Find a function $f(n)$ that identifies the n th term a_n of the following recursively defined sequences, as $a_n = f(n)$.

14. $a_1 = 1$ and $a_{n+1} = -a_n$ for $n \geq 1$

15. $a_1 = 2$ and $a_{n+1} = 2a_n$ for $n \geq 1$
16. $a_1 = 1$ and $a_{n+1} = (n+1)a_n$ for $n \geq 1$
17. $a_1 = 2$ and $a_{n+1} = (n+1)a_n/2$ for $n \geq 1$
18. $a_1 = 1$ and $a_{n+1} = a_n/2^n$ for $n \geq 1$

Plot the first N terms of each sequence. State whether the graphical evidence suggests that the sequence converges or diverges.

19. [T] $a_1 = 1$, $a_2 = 2$, and for $n \geq 3$, $a_n = \frac{1}{2}(a_{n-1} + a_{n-2})$; $N = 30$
20. [T] $a_1 = 1$, $a_2 = 2$, $a_3 = 3$ and for $n \geq 4$, $a_n = \frac{1}{3}(a_{n-1} + a_{n-2} + a_{n-3})$, $N = 30$
21. [T] $a_1 = 1$, $a_2 = 2$, and for $n \geq 3$, $a_n = \sqrt{a_{n-1}a_{n-2}}$; $N = 30$
22. [T] $a_1 = 1$, $a_2 = 2$, $a_3 = 3$, and for $n \geq 4$, $a_n = \sqrt[3]{a_{n-1}a_{n-2}a_{n-3}}$; $N = 30$

Suppose that $\lim_{n \rightarrow \infty} a_n = 1$, $\lim_{n \rightarrow \infty} b_n = -1$, and $0 < -b_n < a_n$ for all n . Evaluate each of the following limits, or state that the limit does not exist, or state that there is not enough information to determine whether the limit exists.

23. $\lim_{n \rightarrow \infty} (3a_n - 4b_n)$
24. $\lim_{n \rightarrow \infty} (\frac{1}{2}b_n - \frac{1}{2}a_n)$
25. $\lim_{n \rightarrow \infty} \frac{a_n + b_n}{a_n - b_n}$
26. $\lim_{n \rightarrow \infty} \frac{a_n - b_n}{a_n + b_n}$

Find the limit of each of the following sequences, using L'Hôpital's rule when appropriate.

27. $\frac{n^2}{2^n}$
28. $\frac{(n-1)^2}{(n+1)^2}$
29. $\frac{\sqrt{n}}{\sqrt{n+1}}$
30. $n^{1/n}$ (Hint: $n^{1/n} = e^{\frac{1}{n} \ln n}$)

For each of the following sequences, whose n th terms are indicated, state whether the sequence is bounded and whether it is eventually monotone, increasing, or decreasing.

31. $n/2^n$, $n \geq 2$
32. $\ln(1 + \frac{1}{n})$
33. $\sin n$
34. $\cos(n^2)$
35. $n^{1/n}$, $n \geq 3$
36. $n^{-1/n}$, $n \geq 3$
37. $\tan n$

38. Determine whether the sequence defined as follows has a limit. If it does, find the limit.
 $a_1 = \sqrt{2}$, $a_2 = \sqrt{2\sqrt{2}}$, $a_3 = \sqrt{2\sqrt{2\sqrt{2}}}$ etc.
39. Determine whether the sequence defined as follows has a limit. If it does, find the limit. $a_1 = 3$,
 $a_n = \sqrt{2a_{n-1}}$, $n = 2, 3, \dots$

Use the Squeeze Theorem to find the limit of each of the following sequences.

40. $n \sin(1/n)$
41. $\frac{\cos(1/n) - 1}{1/n}$
42. $a_n = \frac{n!}{n^n}$
43. $a_n = \sin n \sin(1/n)$

For the following sequences, plot the first 25 terms of the sequence and state whether the graphical evidence suggests that the sequence converges or diverges.

44. [T] $a_n = \sin n$
45. [T] $a_n = \cos n$

Determine the limit of the sequence or show that the sequence diverges. If it converges, find its limit.

46. $a_n = \tan^{-1}(n^2)$
47. $a_n = (2n)^{1/n} - n^{1/n}$
48. $a_n = \frac{\ln(n^2)}{\ln(2n)}$
49. $a_n = \left(1 - \frac{2}{n}\right)^n$
50. $a_n = \ln\left(\frac{n+2}{n^2-3}\right)$
51. $a_n = \frac{2^n + 3^n}{4^n}$
52. $a_n = \frac{(1000)^n}{n!}$
53. $a_n = \frac{(n!)^2}{(2n)!}$

Newton's method seeks to approximate a solution $f(x) = 0$ that starts with an initial approximation x_0 and successively defines a sequence $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$. For the given choice of f and x_0 , write out the formula for x_{n+1} . If the sequence appears to converge, give an exact formula for the solution x , then identify the limit x accurate to four decimal places and the smallest n such that x_n agrees with x up to four decimal places.

54. [T] $f(x) = x^2 - 2$, $x_0 = 1$
55. [T] $f(x) = (x - 1)^2 - 2$, $x_0 = 2$
56. [T] $f(x) = e^x - 2$, $x_0 = 1$
57. [T] $f(x) = \ln x - 1$, $x_0 = 2$

58. [T] Suppose you start with one liter of vinegar and repeatedly remove 0.1 L, replace with water, mix, and repeat.
- Find a formula for the concentration after n steps.
 - After how many steps does the mixture contain less than 10% vinegar?
59. [T] A lake initially contains 2000 fish. Suppose that in the absence of predators or other causes of removal, the fish population increases by 6% each month. However, factoring in all causes, 150 fish are lost each month.
- Explain why the fish population after n months is modeled by $P_n = 1.06P_{n-1} - 150$ with $P_0 = 2000$.
 - How many fish will be in the pond after one year?
60. [T] A bank account earns 5% interest compounded monthly. Suppose that \$1000 is initially deposited into the account, but that \$10 is withdrawn each month.
- Show that the amount in the account after n months is $A_n = (1 + .05/12)A_{n-1} - 10$; $A_0 = 1000$.
 - How much money will be in the account after 1 year?
 - Is the amount increasing or decreasing?
 - Suppose that instead of \$10, a fixed amount d dollars is withdrawn each month. Find a value of d such that the amount in the account after each month remains \$1000.
 - What happens if d is greater than this amount?
61. [T] A student takes out a college loan of \$10,000 at an annual percentage rate of 6%, compounded monthly.
- If the student makes payments of \$100 per month, how much does the student owe after 12 months?
 - After how many months will the loan be paid off?
62. [T] Consider a sequence combining geometric growth and arithmetic decrease. Let $a_1 = 1$. Fix $a > 1$ and $0 < b < a$. Set $a_{n+1} = aa_n - b$. Find a formula for a_{n+1} in terms of a^n , a , and b and a relationship between a and b such that a_n converges.
63. [T] The binary representation $x = 0.b_1b_2b_3 \dots$ of a number x between 0 and 1 can be defined as follows. Let $b_1 = 0$ if $x < 1/2$ and $b_1 = 1$ if $1/2 \leq x < 1$. Let $x_1 = 2x - b_1$. Let $b_2 = 0$ if $x_1 < 1/2$ and $b_2 = 1$ if $1/2 \leq x_1 < 1$. Let $x_2 = 2x_1 - b_2$ and, in general, $x_n = 2x_{n-1} - b_n$ and $b_{n+1} = 0$ if $x_n < 1/2$ and $b_{n+1} = 1$ if $1/2 \leq x_n < 1$. Find the binary expansion of $1/3$.
64. [T] To find an approximation for π , set $a_0 = \sqrt{2+1}$, $a_1 = \sqrt{2+a_0}$, and, in general, $a_{n+1} = \sqrt{2+a_n}$. Finally, set $p_n = 3 \cdot 2^{n+1} \sqrt{2-a_n}$. Find the first ten terms of p_n and compare the values to π .

For the following two exercises, assume that you have access to a computer program or Internet source that can generate a list of zeros and ones of any desired length. Pseudorandom number generators (PRNGs) play an important role in simulating random noise in physical systems by creating sequences of zeros and ones that appear like the result of flipping a coin repeatedly. One of the simplest types of PRNGs recursively defines a random-looking sequence of N integers $a_1, a_2, \dots, a_N$ by fixing two

special integers K and M and letting a_{n+1} be the remainder after dividing Ka_n by M , then creates a bit sequence of zeros and ones whose n th term b_n is equal to one if a_n is odd and equal to zero if a_n is even. If the bits b_n are pseudorandom, then the behavior of their average $(b_1 + b_2 + \cdots + b_N)/N$ should be similar to behavior of averages of truly randomly generated bits.

65. [T] Starting with $K = 16,807$ and $M = 2,147,483,647$, using ten different starting values of a_1 , compute sequences of bits b_n up to $n = 1000$, and compare their averages to ten such sequences generated by a random bit generator.
66. [T] Find the first 1000 digits of π using either a computer program or Internet resource. Create a bit sequence b_n by letting $b_n = 1$ if the n th digit of π is odd and $b_n = 0$ if the n th digit of π is even. Compute the average value of b_n and the average value of $d_n = |b_{n+1} - b_n|$, $n = 1, \dots, 999$. Does the sequence b_n appear random? Do the differences between successive elements of b_n appear random?